Centre For Cybersecurity Institute

Network Remote

Control and Monitoring

Network Remote Control and Monitoring

Scope 1 (Automated): Creating automation that would let cyber units run script from their local devices but would be executed by the remote server. Communicating with a remote server and executing automatic tasks anonymously

Scope 2 (Manual): To better understand network and security, you need to capture and monitor the traffic during the automated attack on the server. Analyse and explain at least 1 unsecure protocol used (telnet or http or ftp), how it impacts the CIA Triad and provide secure alternatives with demonstration.

Project Structure:

Scope 1

1. Installations and Anonymity Check.
   i) Install the needed applications.
   ii) If the applications are already installed. If installed, don't install them again.
   iii) Check if the network connection is anonymous; if not, alert the user and exit.
   iv) If the network connection is anonymous, display the spoofed country name.
   v) Allow the user to specify the address/URL to whois from remote server; save into a variable
2. Automatically Scan the Remote Server for open ports.
   i) Connect to the Remote Server via SSH
   ii) Display the details of the remote server (country, IP, and Uptime)
   iii) Get the remote server to check the Whois of the given address/URL
3. Results
   i) Save the Whois data into file on the remote computer
   ii) Collect the file from the remote computer via FTP or HTTP or any other unsecure protocols.
   iii) Create a log and audit your data collection.
   iv) The log needs to be saved on the local machine.
4. Creativity

General Tools available: sshpass, ssh, nipe, nmap, whois, geoiplookup,

Scope 2: Collecting Information

1. Capture Network Traffic

   i) Use *tcpdump* to capture the network on the Remote Server as you run the automated script

   ii) Save the output into a .pcap file

   iii) Analyse the pcap file with Wireshark

2. Research the selected protocol (Telnet or FTP or HTTP) thoroughly.

   i) Understand its purpose, key features, and the problem it aims to solve. Gather resources such as RFCs (Request for Comments), protocol specifications, and relevant literature.

   ii) Describe the fundamental behaviour of the protocol. Explain how it works, the message exchange process, and the sequence of events. Use diagrams and flowcharts to illustrate the protocol's behaviour.

   iii) Dive deeper into the protocol's mechanisms. Explore aspects like header structure, message formats, and any flags or options that affect its behaviour.

   iv) Discuss the strengths and weaknesses of the protocol and relating it impacting the CIA Triad.

3. Secure Network Protocol

   a) Suggest respective the secure protocol (SSH or SFTP or HTTPS), create a basic client-server application that uses the protocol to exchange data and demonstrate the protocol's behaviour.

   b) Explain how the secure protocol will resolve the impact of CIA triad in part 2iv

4. Conclusion

   i) Conclude the project with a reflection on what you've learned. Discuss how the analysis deepened your understanding of networking protocols and their importance.

General Tools available: tcpdump, wireshark

Report Format:

The report should include at least the following sections and written in formal language:

- Title page
    - The title of the project, your name, student code, the unit code, trainer name
- Table of contents
- Introduction
    - The Introduction section gives a summary of the full report, places your research in context, and emphasizes its importance. It should contain details on the report's purpose, aim, and what will be covered in the rest of the report.
- Methodologies
    - The methodology section describes and explains what you do and what are the outcomes of the various portions of the project.
- Discussion
    - The Discussion section evaluate the findings that were described in earlier sections of your report, determining their significance and crafting an argument in favor of your subsequent recommendations sections. This section's ultimate goal is to interpret the research and assess if the issue was addressed.
- Conclusion
    - The Conclusion section restates the issues stated in the body sections and connects them to construct an argument, consolidates your evaluation of your study and data, highlighting any gaps or issues that will be addressed in the Recommendations section.
- Recommendations
    - The Recommendation section allows you to offer suggestions or solutions to the issue(s) raised in the body and/or conclusion of your report.
- References
    - Both a reference list and the proper in-text citations should be included in your report.
    - Consider using IEEE citation.

Project Deliverables:

- Project should be done on a clean and fresh kali linux and another remote server
- For Scope 1, everything other than the user input should be automated.
    - Use functions for the automation.
    - Use comments in your code to explain what you did.
- For Scope 2, include diagrams, tables, screenshot of scripts and output, and references to relevant sources.
    - PDF report should be written in proper English. No singlish/ bullet points.
    - Refer to the section on Report about referencing and credits.
    - Submit a PDF document providing an introduction to the project, the project objectives, methodologies, and outcomes. It should also include your network schema and the network details for Scope 1.
    - Report should explaine the purpose and usage of each command.
    - Include screenshots of the command used or the process that shows the necessary information such as IP address, protocols etc.
- **Submission**
    - Submit one script (.sh) and one pdf report proving the functions work and the analysis of the network protocol.
    - Put the script (.sh) and pdf report into a Zip File.
    - Submit the project to drive
    - File Name: UNIT.STUDENT NUMBER (e.g. CCK1_240101.s5.zip, CCK1_240101.s5.sh, CCK1_240101.s5.pdf etc.)
    - Failure to do so will be taken as no submission and a 0 grade for the project
    - Please check your report for plagiarism. Any form of it will not be accepted.
    - Upload the project to your individual google drive folder
    - Last Submission: 23:59 of the project submission deadline. Folder will be denied access thereafter.